Actuator Modeling and Predictive Coordination for Coupled Maneuvers of a Moving-Mass ROV

Two fixed thrusters ▪ Moving-mass pitch control ▪ Eight-week study

Summary. We propose to study how accurately a predictive controller must represent a moving-mass mechanism to execute coupled underwater maneuvers and recover from disturbances. The low-cost remotely operated vehicle (ROV) has two independent fixed thrusters and a longitudinal moving mass. Depth changes require pitching the body, while advance and yaw share the motors. We will compare a competent cascade and two nonlinear model predictive controllers (NMPC) differing in carriage-response representation, then assess prediction-horizon effects. Simulation proceeds alongside construction; selected pool trials test the identified mechanisms. The intended outcome is evidence about when simple prediction suffices, when additional actuator dynamics help, and where physical limits dominate.

1 Context and control problem

The team's earlier formation studies [1] investigate tracking and separation near operating limits. This project examines the individual vehicle beneath that coordination layer: a follower must execute changing position references and recover when disturbed. One robot provides a manageable control problem while mechanical construction proceeds in parallel. The moving mass replaces the original concept's vertical thruster; this configuration must be agreed with the team.

For longitudinal forces F_L, F_R and thruster half-spacing b, the ideal propulsion geometry gives $T = F_L + F_R$ and $\tau_z = b(F_R - F_L)$, up to sign convention. With symmetric force bounds, $|T| + |\tau_z|/b \leq 2F_{\max}$. Turning thus consumes capacity also needed for translation. The vertical thrust component depends on pitch, whose response is limited by carriage travel, velocity and force. A lateral disturbance requires reorientation and motion rather than direct lateral cancellation.

> **When must a controller explicitly predict moving-mass dynamics to track coupled turning–depth maneuvers and recover from disturbances, and when is a simpler actuator approximation sufficient?**

The task is tracking a moving, time-indexed position reference. Attitude supports translation; arbitrary six-degree-of-freedom pose tracking is not commanded. Reference timing and available preview are matched across treatments. Future force pulses are unknown to every controller. Silently slowing a reference would change the task.

2 Literature and proposed contribution

SAM already demonstrates predictive underwater coordination with moving mass and other actuators; its published formulation uses longitudinal center-of-gravity position as a control input [2]. Adaptive MPC for moving-mass depth control already addresses displacement/velocity limits, uncertainty and hysteresis [3]. Direct-thruster learning MPC has also been demonstrated for horizontal-plane tracking [5]. Joint actuation, constraints and prediction are established ingredients.

The candidate contribution is an *isolated comparison of carriage-response models and prediction horizons* under coupled maneuver demands, with calibration data and other control components held comparable. We seek explanatory tracking/recovery maps, computation trade-offs and selected physical confirmation. Changing actuator layout or outperforming PID alone is insufficient. The originality claim requires a complete closest-work check and a repeatable pilot effect; a more detailed model is not presumed superior.

3 Physical model, measurements and calibration

Simulation plant and control predictor

The simulation plant will combine an articulated MuJoCo body and carriage with marine forces: gravity, buoyancy, hydrodynamic damping and effective inertia. Published BlueROV2 modeling [9] provides a reference for terms and initial coefficients; a different hull requires its own checks. This is a vehicle-dynamics approximation, without resolving the surrounding flow through computational fluid dynamics.

The controller will use a smaller, differentiable predictor based on marine-vehicle and moving-mass mechanics [6]. Moving the mass changes the center of gravity and restoring moments. Use the same hull, thruster and mass-coupling formulation in both NMPC variants, retaining reaction terms when relevant. Their intended difference is carriage response: a simple bounded approximation versus a dynamic servo model supported by measurements. Position, orientation, body velocities and actuator feedback initialize prediction; the richer variant also propagates carriage velocity explicitly.

Measurements available on the real vehicle

Intended inputs are pose from overhead and side cameras, IMU data and carriage feedback. Velocity will be estimated causally; unmeasured motor states may be propagated from command history. Camera calibration, refraction, timestamp alignment and delay must be measured. Simulation observations will reproduce the relevant errors and update rates. Exact fluid forces and simulator-only velocities are excluded from controller inputs. The proposed camera setup supplies localization, without requiring learned perception or obstacle detection.

A bounded calibration task

First measure total mass, geometry, travel, buoyancy and balance. Check that mass leverage and carriage speed permit the proposed maneuvers; an inadequate pitch mechanism cannot be repaired by control software. Short carriage steps and simple propulsion/pitch runs estimate servo response, thrust response and dominant damping. Use documented priors for parameters that limited data cannot identify reliably.

Fit both carriage approximations from the same runs and freeze common hull parameters. Reserve whole maneuvers and sessions for prediction checks, then freeze fitted models before control evaluation. Calibration quality and data acquisition protocols are not additional experimental factors. Before hardware exists, use nominal parameters and reserved uncertainty settings. Report sensitivity to calibration and prediction mismatch. Adopt extra servo states only if response data justify them; continuous parameter learning remains optional.

4 Predictive control and matched disturbance treatment

Nonlinear model predictive control repeatedly optimizes a future sequence of commands using

$$\hat{x}_{k+1} = f(\hat{x}_k, u_k, \hat{d}_k; \theta), \qquad u_k = [u_{L,k}, u_{R,k}, s_k^{\mathrm{ref}}]^{\mathsf{T}}, \tag{1}$$

where θ contains fitted model parameters. Both variants minimize position-tracking error and command variation with matched physical limits. At each update, estimate the current state, optimize a future command sequence and *apply only the first command*. Repeat with new measurements. This requires online optimization, without policy training. The motor commands and carriage reference act through their modeled responses. A depth reversal during a turn tests the value of preparing the required thrust direction.

Choose a common initial horizon using measured carriage/body response. CasADi/acados supplies optimization infrastructure. Record solver latency, missed deadlines and fallback use; modeled constraints do not guarantee real feasibility under arbitrary mismatch. A limited horizon study follows nominal model comparisons.

Start without bias estimation in nominal tests. For persistent-disturbance cases, use the same small observer design [4], causal measurements and feasible compensation principles across controllers. Effective force/moment biases are updated from residuals and held constant within each horizon. They can include model error and do not measure current directly. Observer removal is a secondary check on selected cases, rather than a second full factorial experiment. No controller receives future pushes or privileged fluid forces.

5 Controller comparisons and prediction horizon

Three treatments with an interpretable internal comparison

The cascade uses filtered guidance, suitable speed/attitude PID loops, feedforward, anti-windup and feasible motor allocation. It accounts for carriage limits and receives the same reference information and measurements. Tune it on the nominal development set with comparable effort. PID performance may be sufficient; the study must allow that outcome.

Treatment	Carriage representation	Purpose
A: Cascade	Feedback with measured limits and response-aware references	Practical value of predictive control
B: Simple NMPC	Fitted first-order or kinematic response with speed/travel limits	Value of inexpensive prediction
C: Dynamic NMPC	Measured servo dynamics; propagate carriage position and velocity	Added value of response fidelity

For B versus C, hold hull physics, thruster response, cost, nominal horizon and shared constraints fixed. Fit from identical data and use equal tuning budgets. An instantaneous-carriage model may be a diagnostic, but is not a credible main competitor. Keep mass-reaction terms identical when isolating servo response. Do not infer that C is needed merely because it has more states.

A limited anticipation study

On selected cases, compare a few horizons shorter than, near and longer than measured carriage/body preparation time. This time includes pitch response, not just rail traversal, and depends on the maneuver and starting state. Keep sampling period and reference information fixed, report solver cost and missed deadlines, and retain model mismatch. Horizon length is an experimental variable, not a guaranteed recovery condition. Avoid a full cross-product of model, horizon and observer choices.

6 Simulation experiments and evidence of contribution

Separate the demands. Start with a depth reversal during straight advance, a constant-depth turn, and the depth transition performed during a turn. Match component profiles where possible and vary their coincidence and speed. A gentle helix is an additional demonstration, not the main mechanism test. Use physically coherent initial carriage/body states. Explore easy, intermediate and demanding regimes.

Nominal tracking and recovery. First map nominal tracking performance. Then apply unknown force pulses in a common domain where all treatments were tracking acceptably before the pulse. Explore onset phase, direction and magnitude. Failure to track before a push is not classified as failed recovery. Constant currents and reserved parameter/sensor errors test robustness subsequently; time-varying flow needs a verified extension.

Frozen, paired evaluation. Freeze models, controller tuning, error tolerance, recovery deadline and dwell interval after the pilot. Reserve whole trajectories and uncertainty combinations; distinguish plant and predictor beyond integration details. Use at least ten paired realizations per selected simulation condition, retain failures and report trial-level uncertainty. Report tracking error/lag, recovery success/time, saturation and computation. Carriage, pitch and individual-thrust traces must explain differences; maps are empirical, not optimal reachable sets.

Tests of the mechanism. Test whether added response fidelity changes performance preferentially during simultaneous pitch reversal and turning, and whether that effect depends on preparation time relative to prediction horizon. Check whether benefits survive held-out mismatch and computation deadlines. Simpler prediction may suffice; similar errors alone do not prove equivalence. A useful contribution needs repeatable effects or well-supported practical sufficiency, beyond an NMPC-versus-PID ranking.

7 Pool validation, delivery and hardware independence

A small physical validation with explicit claims

Test two representative conditions: one where simple prediction is expected to suffice and one where the pilot predicts a response-model effect. Aim for five trials per treatment and condition, rotating order; include B and C to test the mechanism rather than only A versus C. Freeze the protocol before confirmation. Camera/IMU logs capture tracking and pull-and-release recovery. Match measured post-release errors, velocities and carriage states; equal-looking pulls do not establish equal impulses without force measurements.

The controller can run on an external computer through the tether. Confirm carriage feedback, actuator interfaces, localization/refraction and timing early. Direct motor commands and a carriage reference are the proposed interface; any onboard firmware must expose that interface or be included consistently in the model and both controllers. The project does not assume that a particular autopilot is already installed. Record disturbances from the hardware tether as a possible confound.

Published model components and numerical checks support simulator construction; they do not validate our custom vehicle. Compare held-out physical motion with predictions and report the remaining mismatch. If hardware is delayed, simulation still supports the course deliverable and a reproducible control study, with claims restricted to simulation. Real transfer claims require pool evidence.

Eight-week plan and decision gates

Period	Deliverable and decision
Weeks 1–2	Realistic observations, competent cascade and simple NMPC. Check feasible maneuvers, solver timing and carriage-response assumptions.
Weeks 3–4	B/C model comparison, targeted horizon pilot, first hardware logs and shared-data fitting. Look for an explanatory effect before adding methods.
Weeks 5–6	Freeze models and tuning; paired held-out simulation and pool tests. Confirm the mechanism under mismatch; extensions follow core evidence.
Weeks 7–8	Paired analysis, limitations, reproducible release and paper writing. Optional results remain separately labeled.

One software lead develops control while the remaining team builds the ROV; aim for a first pool check before 7 November. If the pilot shows no meaningful model/horizon effect, simplify the controller and revise the publication claim. If gains vanish under small mismatch, investigate that sensitivity. Diagnose physical infeasibility, model error and solver failures before adding complexity. A stable simulator and baseline preserve the course deliverable.

8 Staged research extensions

Alternative control or learning. MPPI [7] is a technical alternative if useful differentiable prediction or NMPC optimization becomes impractical; it still needs a predictor and rollout computation. PPO has underwater transfer precedent [8], but adds training, multiple seeds and a generalization study. Consider it after the core task is stable, to test feedback coordination or online computation. Learned residuals or robust MPC require a diagnosed model deficit or specified uncertainty question. None is a required fourth benchmark or a remedy for weak evidence.

Return to formations. The individual-vehicle controller can later execute references from a formation layer, connecting back to the team's earlier studies. A learned shared policy could instead use proprioception, relative neighbor motion and desired geometric relationships. Three-dimensional geometry can encode depth differences, but does not remove sensing or actuation limits. Leader-free coordination and freedom to exchange formation slots are separate design choices. Start with two vehicles and explicit targets; role exchange, variable team size and a GNN require additional questions and experiments. Shared weights alone do not establish novelty or generalization.

Priority. Establish the single-vehicle mechanism before attempting cooperation. The two-ROV study below is a connected future direction, turning another vehicle into a structured load. It does not substitute for an inconclusive core experiment, and completion is not required within eight weeks.

9 Optional extension: two ROVs connected by a flexible line

Control question. Connect two simulated copies with a passive line and track a common moving path while keeping tension in a modest prescribed band. Each robot then influences the load and motion of the other. This creates a coupled task beyond following two unrelated points. Marine cable-connected cooperation already has precedents, including surface–underwater systems [10]; a new two-ROV demonstration alone would not establish novelty.

Objective and initial model. Coordinate pair motion, separation and tension while respecting both vehicles' motor and carriage limits. Avoid maximizing tension: excessive pulling consumes authority and can amplify disturbances. Begin with a short, near-neutrally buoyant line and a tension-only compliant approximation that permits slack. Check equal-and-opposite attachment forces, attachment torques, slack-to-taut transitions and integration sensitivity. This approximation omits distributed cable drag and shape; use a segmented model if those effects alter the conclusions. Endpoint distance alone cannot identify tension in a general flexible cable.

First controller and experiment. Start with centralized joint NMPC, receiving both vehicle states and optimizing four motor commands and two carriage references with a line-load predictor. Compare against two independently tracking controllers. Use feasible pair trajectories, then perturb one vehicle or vary its carriage response. Measure pair tracking, peak tension, time outside the tension band, slack/retension and disturbance transmission. Test whether anticipating the partner's response improves load regulation and recovery. This extends the core question to mechanically transmitted disturbances.

Scope and measurements. Start in simulation with known line properties and state uncertainty. If tension is observed, declare it as a simulated force-sensor assumption and test noise/delay; hardware tension feedback would need measurement or validated estimation. Decentralized communication, learned cooperation and a second physical robot are subsequent steps. This extension deserves its own modeling checks and literature review before making a specific originality claim.

References

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